

Tokenization Multiplicity Leads to Arbitrary Price Variation in LLM-as-a-service

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LLMs can generate the same output with different tokenizations

You are a helpful assistant that rephrases text. Provide only the rephrased text.
Liste des récompenses et nominations de Eddie Murphy, notamment pour ses performances d'acteur et musicales.

Énumération des distinctions et nominations d'Eddie Murphy, en particulier pour ses performances d'acteur et musicales.

Canonical tokenization, seen during training

23 tokens

Énumération des distinctions et nominations d'Eddie Murphy, en particulier pour ses performances d'acteur et musicales.

Non-canonical tokenization, can be generated at inference

24 tokens

The problem: Same input Same output Different price

Code



Data



Empirical study

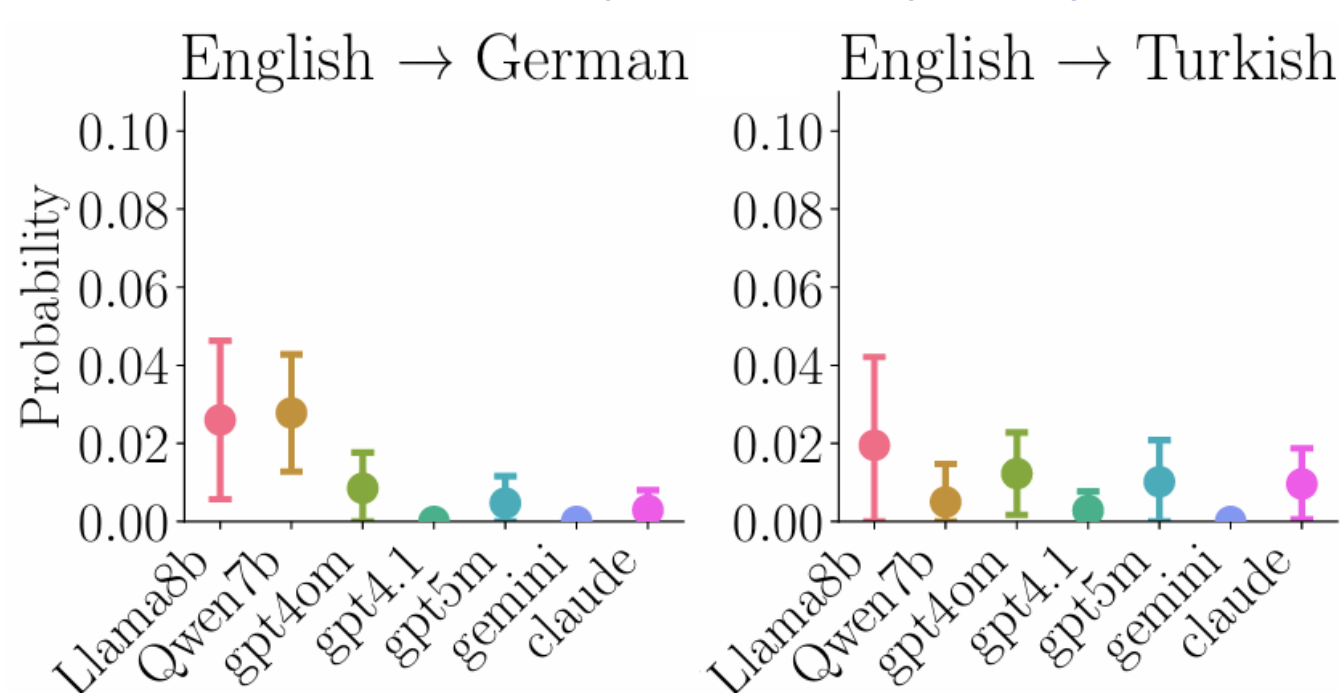
We prompted LLMs with the same inputs 100 times to study this price variation.

3 realistic tasks involving processing 100 small Wikipedia articles in 6 languages:

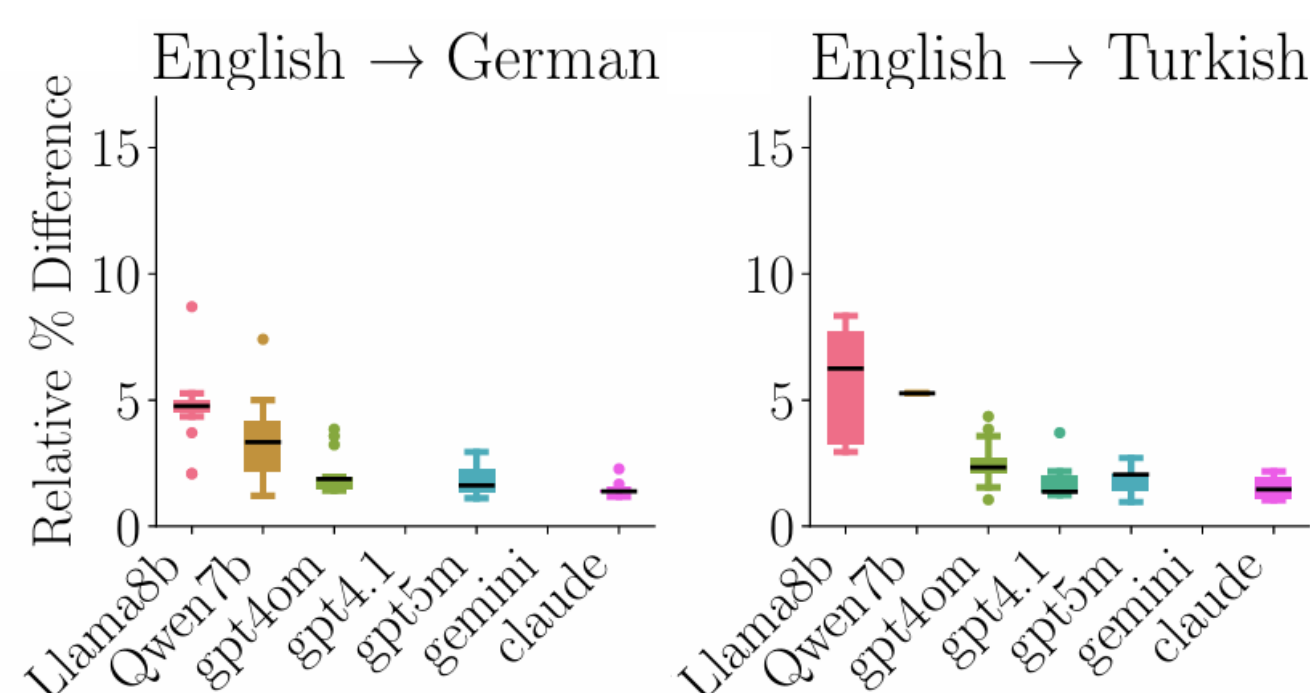
- Translation
- Rephrasing
- Spell Checking

Example results: translation from English to German and Turkish

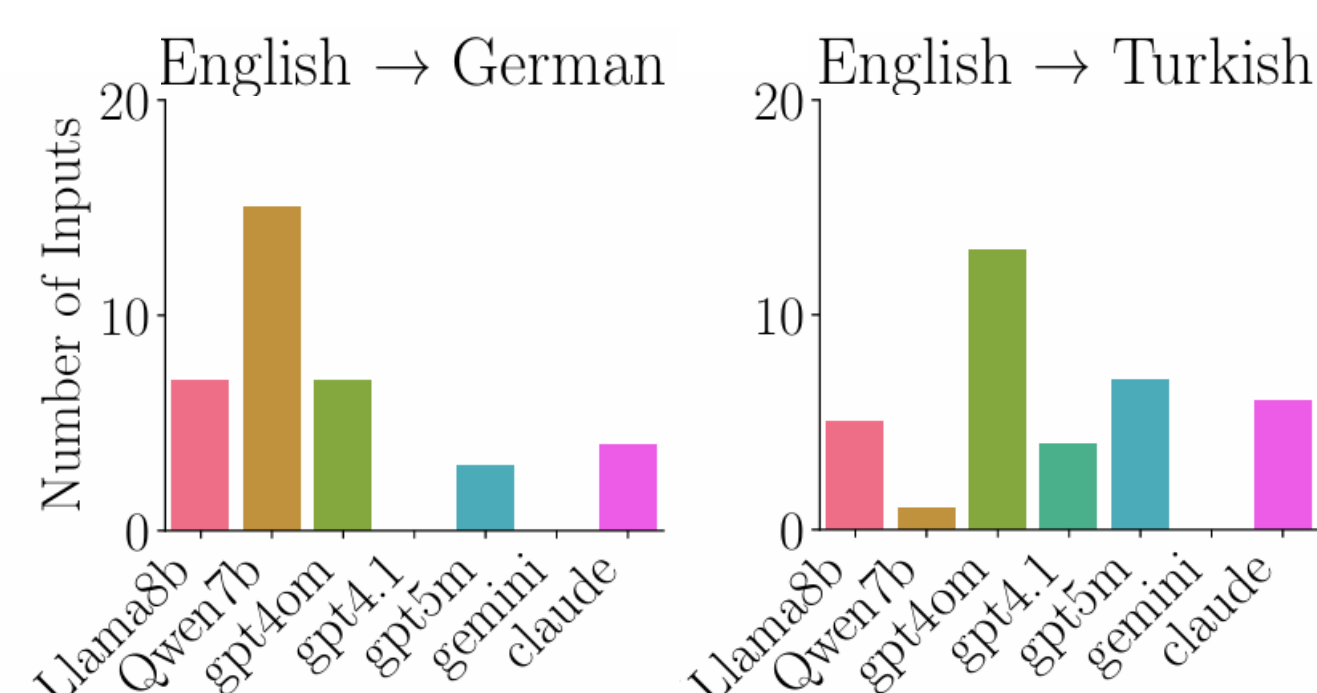
Probability of multiplicity



Magnitude of price variation

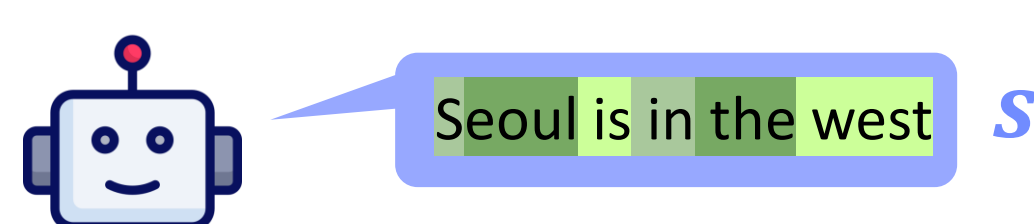


Occurences of multiplicity



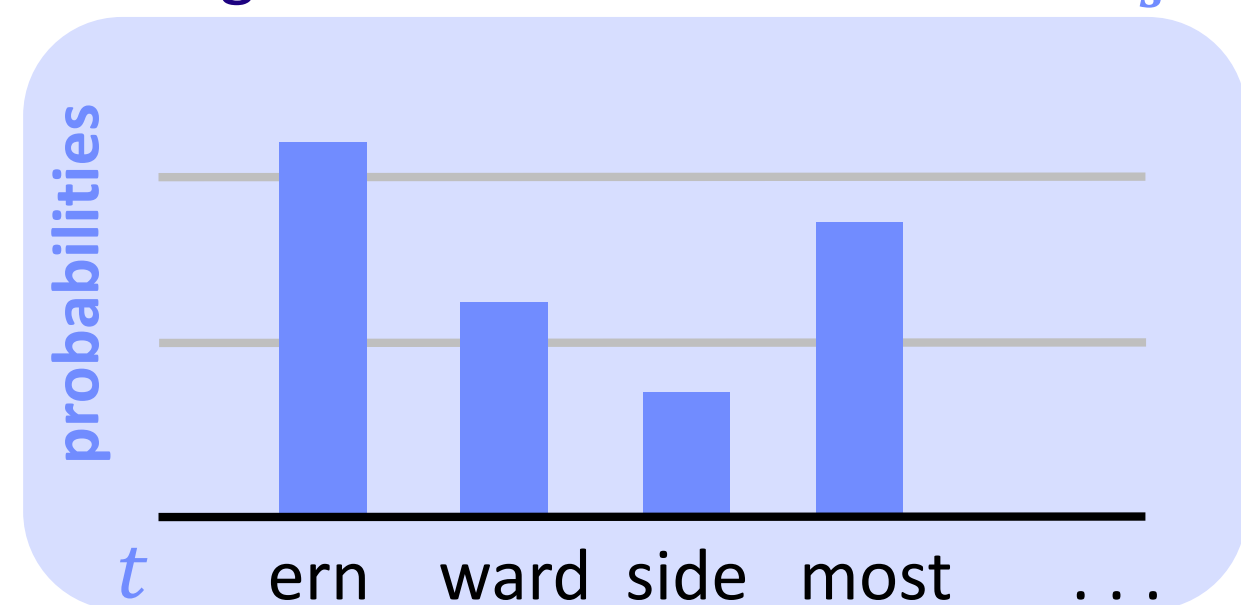
Our solution: Canonical Autoregressive Generation

Theorem: at every step of generation the (partial) token sequence must be canonical



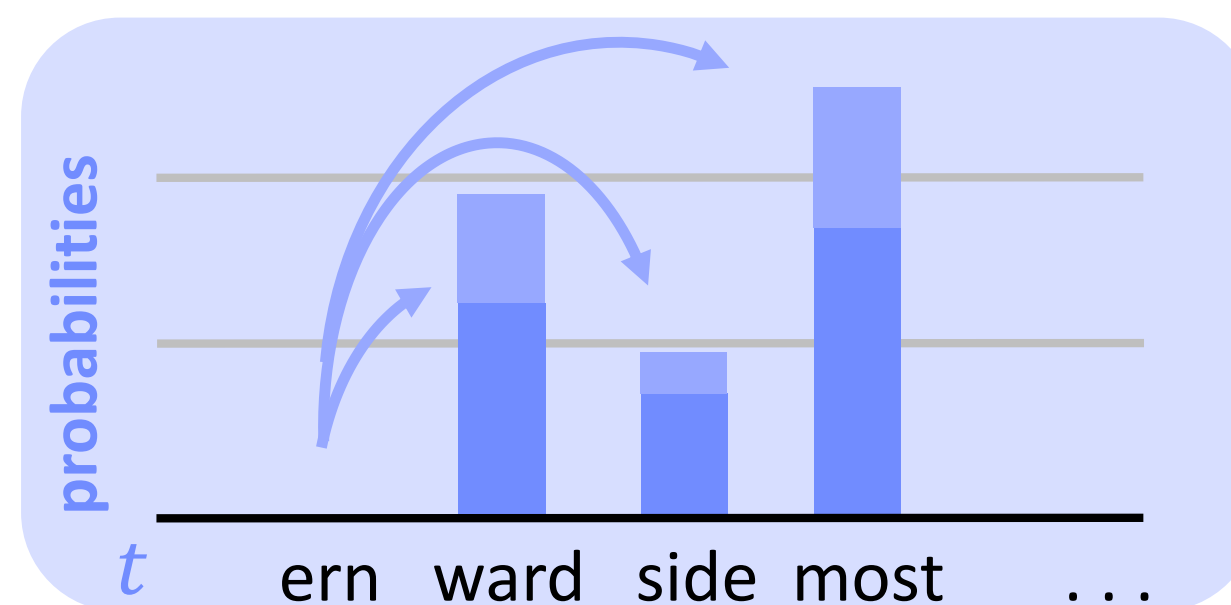
Restrict generation to canonical tokenizations, for deterministic tokenization (and price) per string.

Original next-token distribution d_s



$s|t$ is canonical ✗ ✓ ✓ ✓

Canonicalized next-token distribution $\tilde{d}_s(t)$



$$\tilde{d}_s(t) = \begin{cases} \frac{d_s(t)}{Z} & , s | t \text{ is canonical} \\ 0 & , s | t \text{ is non-canonical} \end{cases}$$

Theorem:

Sampling from $\tilde{d}_s$ leads to output token sequences closer to the true distribution of token sequences (as seen during training)

Computing the normalization constant $Z = \sum_{t \in V: s|t \text{ is canonical}} d_s(t)$ requires checking if $s|t$ is canonical for all tokens t

But we can efficiently sample from $\tilde{d}_s$ using the **Gumbel-Max trick**:

$$\tilde{d}_s(t) \sim \operatorname{argmax}_{t \in V: s|t \text{ is canonical}} \{\log(d_s(t)) + u_t\}$$

ALGORITHM

Sample $u_t \sim \text{Gumbel}(0,1)$ for every token t in the vocabulary

For every token t in decreasing order of $\log(d_s(t)) + u_t$:
If $s|t$ is canonical then return t

Requires fewer than $\frac{1}{Z}$ canonicity checks on average

Paper



Performance & Runtime

Task	Metric	Llama8B	
		Standard	Canonical
Translation	Quality score	0.72 ± 0.02	0.70 ± 0.02
	Time per token (s)	0.019	0.020
	Non-canonicity rate	6%	-
Spell Checking	1 – edit distance	0.62 ± 0.04	0.61 ± 0.04
	Time per token (s)	0.020	0.023
	Non-canonicity rate	10%	-
Rephrasing	Cosine similarity	0.84 ± 0.02	0.84 ± 0.02
	Time per token (s)	0.020	0.020
	Non-canonicity rate	6%	-
MGSM	Accuracy	0.37 ± 0.06	0.37 ± 0.06
	Time per token (s)	0.020	0.021
	Non-canonicity rate	22%	-